Identifying Colors of Products and Associated Personalized Recommendation Engine in e-Fashion Business

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Abstract

One of the important factors ignored in the literature in e-marketing is "the color" of a product. While one may be able to identify the dominating color of a product based on the overall impression, it is not easy to mechanize the process to determine the dominating color. Accordingly, in many applications, the color of a product is defined subjectively by those who enter the data. Consequently, the color of a product has been a missing link in e-marketing. The purpose of this research is to fill this gap by developing an algorithmic procedure for identifying the dominating color of a product by analyzing a digital image of the product. The algorithmic procedure enables one to reveal color preferences of consumers by analyzing the digital images of the products obtained from the purchasing records. A recommendation engine is also developed based on color class preference vectors of individual consumers.

Keyword: e-Marketing, Product color, Identification algorithm, Color class preference vector

1 Identifying Colors of Products

In recent years, as pointed in [1], there has been an increasing interest in Web usage mining as a means to capture Web user behavioral patterns and to derive e-business intelligence. In [2, 3], for example, automatic personalization was proposed based on clustering of user transaction and page-views. To the best knowledge of the authors, however, the color information has not been incorporated in the literature for developing better personalized recommendation engines. This is so because it is difficult to define a color of a product, which typically consists of many different colors. While one dominating color may be identified in the eye of human, it is not easy to mechanize the process to determine the dominating color. Accordingly, in many applications, the color of a product is determined subjectively by those who enter the data. Furthermore, terms for describing a color are often quite vague and too many. Because of these reasons, the color of a product has been a missing link in e-marketing. This research fills this gap by developing an algorithmic procedure for identifying the dominating color of a product by analyzing a digital image of the product.

To identify the colors of products, we determine a vector $\underline{\Phi}(v_p)$ in CIE-L*a*b*, which we call a CCPV (Color-Class Profile Vector) of a digital image v containing a product p. Each component was determined as the Euclidean distances between each of the transformed vectors and six fixed points in CIE-L*a*b* representing Red, Green, Yellow, Blue, White and Black and then taking the average over the pixels in the digital image v_p. Since each product has its color in the data base, these vectors can be utilized to categorize many different colors, resulting in 14 color classes. This enables one to study color preferences of consumers by segments. Furthermore, it provides a basis for constructing a recommendation engine based on the color classes for enhancing e-commerce.

2 Numerical Experiments of Recommendation Engine

To validate the personalized recommendation by applying the algorithmic procedure to a set of digital images of products purchased by a consumer, the numerical experiments based on real data was conducted. The data was obtained from a corporation, which has been in retail business worldwide, offering a variety of products ranging from Apparel products, Jewelries, and Home electronics appliances to foods. It consist of demographic information of those consumers who purchased at least one product during the period between September 1st, 2004 and August 31st, 2007. The amount of consumers was about 455 thousand, the amount of product categories was 34 and the data consisted of about 2.3 million records.

As the results of numerical experiment based on cross validation approach, the accuracy of the recommendation engine on the proposed method was about 1.5 times greater than it of the conventional method, which ignored the colors of the products, for almost all of the categories of the product.

References

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